

Sarthak Kumar Maharana

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Research Interests

Short summary: Computer Vision and Machine Learning. Core interests: Continual Learning, Multimodal Learning, Adaptive and On-Device ML

Education

- 2023-Present **PhD in Computer Science**, *The University of Texas at Dallas*, Richardson, USA, 3.85/4.0
Advisor: Yunhui Guo; Expected graduation: 05/27
- 2021-2023 **MS in Electrical Engineering**, *University of Southern California (USC)*, Los Angeles, USA, 3.85/4.0
Track: Machine Learning and Data Science.
- 2016-2020 **BTech in Electrical and Electronics Engineering**, *International Institute of Information Technology (IIIT)*, Bhubaneswar, India, 8.32/10, Thesis: 9.85/10.0
Thesis: *Acoustic-to-articulatory inversion of dysarthric speech by using cross-corpus acoustic-articulatory data.*

Experiences

- May'26- **NEC Laboratories America, Inc.**, San Jose, USA
Present *Research Intern* — Media Analytics Group; Mentors: Yi-Wen Chen, Francesco Pittaluga, Vijay Kumar BG, Manmohan Chandraker
○ Working on multimodal planning and reasoning for autonomous driving in long-tailed scenarios.
- May'25- **Dolby Laboratories**, Sunnyvale, USA
Aug'25 *PhD Research Intern* — Sight Experiences Lab (Advanced Technology Group); Mentors: Akshay Mehra, Bhavya Ramakrishna, Guan-Ming Su
○ Proposed a novel model-agnostic inference-time framework for robust continual audio-visual learning.
- Aug'23- **Data-Efficient Intelligent Learning Lab**, *UTD*, Richardson, USA
Research Assistant — Advisor: Yunhui Guo
○ Working on continual learning schemes for long-context modeling; generation for customization and personalization.
○ Designed efficient continual test-time model adaptation schemes to enhance robustness to out-of-distribution data.
- May'22- **USC's Mark and Mary Stevens Neuroimaging and Informatics Institute**, *Neuro Imaging Computing Research*, Los Angeles, USA
July'23 *Student Researcher* — Advisor: Yonggang Shi
○ Developed a tool to perform automatic tractography of the brainstem using d-MRI images.
○ Leveraged image registration and label fusion methods to automatically generate the anatomical ROIs.
- Dec'21- **Signal Analysis and Interpretation Lab (SAIL)**, *USC*, Los Angeles, USA
Dec'22 *Student Researcher* — Advisor: Shrikanth (Shri) Narayanan
○ Speaker recognition from real-time MRI videos, based on an unsupervised disentanglement representation learning scheme.

- Dec'19- **Signal Processing and Interpretation (SPIRE) Lab**, *Indian Institute of Science (IISc)*,
Sep'20 Bengaluru, India
Student Researcher — Advisor: Prasanta Kumar Ghosh
- Studied acoustic-to-articulatory inversion (AAI) for dysarthric speech at low-resource data conditions involving Indian languages, using joint learning and multi-task training. [ICASSP'21]
 - Conditioned the jointly-trained AAI model with x-vectors to study its benefits on the AAI performance of dysarthric subjects.

Publications & Preprints

- * See my Google Scholar for an updated list.
- 1 **Audio-Visual Continual Test-Time Adaptation *without* Forgetting**
Sarthak Kumar Maharana, Akshay Mehra, Bhavya Ramakrishna, Yunhui Guo, and Guan-Ming Su.
ECCV 2026 & ICML Continual Adaptation at Scale: Towards Sustainable AI Workshop 2026
[Paper]
 - 2 **Continual Test-Time Adaptation in Computer Vision: Methods, Benchmarks, and Future Directions**
Sarthak Kumar Maharana*, Shambhavi Mishra*, Yunbei Zhang*, Shuaicheng Niu, Taki Hasan Rafi, Jihun Hamm, Marco Pedersoli, Jose Dolz, and Yunhui Guo.
[Under Review]
[Paper]
 - 3 **Variational Diffusion Unlearning: A Variational Inference Framework for Unlearning in Diffusion Models under Data Constraints**
Subhodip Panda, MS Varun, Shreyans Jain, Sarthak Kumar Maharana, and Prathosh AP.
TMLR 2026 & NeurIPS Safe Generative AI Workshop 2024
[Paper]
 - 4 **AVROBUSTBENCH: Benchmarking the Robustness of Audio-Visual Recognition Models at Test-Time**
Sarthak Kumar Maharana, Saksham Singh Kushwaha, Baoming Zhang, Adrian Rodriguez, Songtao Wei, Yapeng Tian, and Yunhui Guo.
NeurIPS Datasets & Benchmarks Track 2025
[Paper][Code][Datasets][Project][Demo]
 - 5 **BATCLIP: Bimodal Online Test-Time Adaptation for CLIP**
Sarthak Kumar Maharana, Baoming Zhang, Leonid Karlinsky, Rogerio Feris, and Yunhui Guo.
ICCV 2025
[Paper][Code][Project][Presentation]
 - 6 **SELECT: A Submodular Approach for Active LiDAR Semantic Segmentation**
Ruiyu Mao, Sarthak Kumar Maharana, Xulong Tang, and Yunhui Guo.
[Under Review]
 - 7 **PALM: Pushing Adaptive Learning Rate Mechanisms for Continual Test-Time Adaptation**
Sarthak Kumar Maharana, Baoming Zhang, and Yunhui Guo.
AAAI 2025 (Oral)
[Paper][Code][Project][Presentation]
 - 8 **STONE: A Submodular Optimization Framework for Active 3D Object Detection**
Ruiyu Mao, Sarthak Kumar Maharana, Rishabh K Iyer, and Yunhui Guo.
NeurIPS 2024
[Paper][Code]
 - 9 **Not Just Change the Labels, Learn the Features: Watermarking Deep Neural Networks with Multi-View Data**
Yuxuan Li, Sarthak Kumar Maharana, and Yunhui Guo.
ECCV 2024
[Paper] [Code]

- 10 **Acoustic-to-articulatory inversion for dysarthric speech: Are pre-trained self-supervised representations favorable?**
Sarathak Kumar Maharana, Krishna Kamal Adidam, Shoumik Nandi, and Ajitesh Srivastava.
ICASSP Self-supervision in Audio, Speech, and Beyond Workshop 2024
[Paper] [Poster]
- 11 **Acoustic-to-articulatory inversion for dysarthric speech by using cross-corpus acoustic-articulatory data**
Sarathak Kumar Maharana, Aravind Illa, Renuka Mannem, Yamini Belur, Preetie Shetty, Veeramani Preethish Kumar, Seena Vengalil, Kiran Polavarapu, Nalini Atchayaram, and Prasanta Kumar Ghosh.
ICASSP 2021
[Paper] [Poster] [Slides]

Selected Projects

- Oct'22- **Understanding Multi-Modal Speaker Recognition via Disentangled Representation**
Dec'22 **Learning**, USC
- Presented an adversarial invariance approach to address multimodal speaker recognition, robust to various sources of variability present in videos and speech. [Code], [Report]
- Nov'22- **Understanding Linguistic Patterns for Text-Based Speaker Classification**, USC
Dec'22
- Studied various text feature extraction methodologies using pre-trained models and classification algorithms and compared them to build a computationally efficient system targeted for text-based speaker classification. [Code], [Report]
- Apr'22- **The Effect of Conditioning of Trigonometric Transformations of Dates with Meteorological Data in Forest Fires Prediction: An Experimental Study**, USC
May'22
- Studied the effects of conditioning a trigonometric transformation of dates with meteorological data, that would aid in predicting the occurrence of forest fires in Algeria. [Code]
- Jan'19- **Single Image Haze Removal using Dark Channel Prior**, IIIT-Bh
Mar'19
- The dark channel is based on the following observation relating to outdoor images: In most of the non-sky patches, at least a certain color channel has at least one dark channel i.e. it has low intensities tending to zero.
 - Redesigned an algorithm regarding this. [Link].

Skills

Languages	<i>Advanced</i> : Python, MATLAB; <i>Familiar</i> : C++, Java, Bash
ML Libraries	<i>Advanced</i> : PyTorch, Keras, TensorFlow; <i>Intermediate</i> : OpenCV, scikit-learn
Softwares	<i>Intermediate</i> : RStudio, ITK-SNAP, BrainSuite
Others	Git, $\LaTeX$; OS: Unix, Windows

Talks

BATCLIP: Bimodal Online Test-Time Adaptation for CLIP

1st Workshop on Multimodal Continual Learning, ICCV 2025

PALM: Pushing Adaptive Learning Rate Mechanisms for Continual Test-Time Adaptation

AAAI'25 (Oral)

Acoustic-to-articulatory inversion of dysarthric speech by using cross-corpus acoustic-articulatory data

ICASSP'21

Teaching Experience

Computer Vision, *Teaching Assistant*, UTD (Jan'26 - May'26)

Artificial Intelligence, Operating Systems, Teaching Assistant, UTD (Aug'23 - Dec'23)
A Computational Introduction to Deep Learning, Grader and Mentor, USC (Jan'22 - May'22)

Academic Service

Workshop Organizer

- 1st Workshop on Multimodal Continual Learning, ICCV'25
- 2nd Workshop on Test-Time Adaptation: Putting Updates to the Test!, ICML'25

Invited Reviewer

- TMLR'26, NeurIPS'26, CVPR'25, ICLR'25, NeurIPS-W'24, BMVC'24, CVPR-W'24, ECCV'24, AAAI'24.

Awards

- 2025 **Travel grant**, Scholarship offered by the IEEE/CVF society to present at ICCV 2025.
- 2024 **Oxford Summer School in Machine Learning 2024**, Accepted to the MLx Representation Learning and Generative AI track
- 2023 **Computer Science PhD Fellowship**, The University of Texas at Dallas
- 2020 **Governing Body Merit Scholarship**, Award of INR 15k, *Academic year: 2019-2020*, IIIT-Bh
- 2019 **Dean's List** [Link], IIIT-Bh
- 2019 **Summer Research Fellowship**, *Indian Academy of Sciences (10% selection rate)* [Link]

Volunteer Work

- Nov'25- **Comets AI Research (CAIR)** — Working as a research advisor to BS/MS students.
- May'24- **UTD CS K-12 Outreach Program** — Served as a mentor to 5 high school students to
Jul'24 conduct research projects.
- Aug'23- **CORD.ai** — Helping build CORD.ai, an AI research community, as a research advisor and
Aug'24 volunteer.
- Oct'21- **USC IEEE Graduate Society** — Attended group meetings, strengthened academic and
May'23 social growth of the members, and hosted workshops.
- Aug'20- **PyCon India 2020** — Content writer for social media handles, helped the promotions team,
Oct'20 and created virtual swags.
- Jul'18- **International Association of Engineers (IAENG)** — Student member.

Mentoring

Kartik Joshi - UT Dallas BS CS
Harshal Agrawal — UT Dallas MS CS → Data Engineer @ Supreme Lending
Adrian Rodriguez — UT Dallas BS CS → MS CS (Thesis)
Shreyans Jain — IIT Gandhinagar BTech
MS Varun — PES University BTech → Software Engineer @ Visa